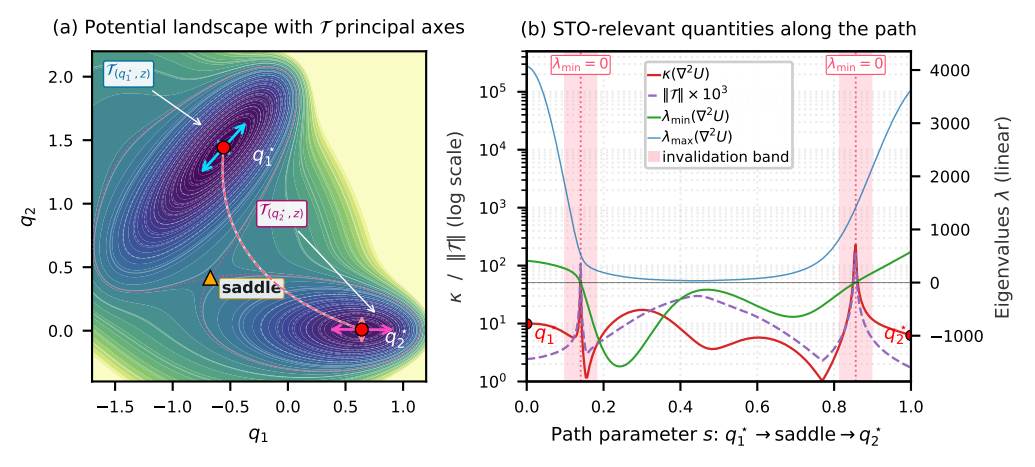

(a) Potential landscape with $\mathcal{T}$ principal axes
$\mathcal{T}_{(q_1^\star, z)}$
$q_1^\star$
$\mathcal{T}_{(q_2^\star, z)}$
saddle
$q_2^\star$
q_2
q_1
(b) STO-relevant quantities along the path
$\lambda_{\min} = 0$
$\lambda_{\min} = 0$
$\kappa(\nabla^2 U)$
$\|\mathcal{T}\| \times 10^3$
$\lambda_{\min}(\nabla^2 U)$
$\lambda_{\max}(\nabla^2 U)$
invalidation band
κ / $\|\mathcal{T}\|$ (log scale)
Eigenvalues λ (linear)
$q_1^\star$
$q_2^\star$
Path parameter s: $q_1^\star \to$ saddle $\to q_2^\star$